ICT2113 Software Modelling and Analysis

Project Specification (PS)

Android-based Biometric Student Attendance System

AY2025/2026, Trimester 2

By: P2-1561W

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BIOSEC

1. Company Description

Company Name: BioSec

Company Address: 50 Nanyang Ave, Singapore 639798

Industry Sector: Education

Line of Business: Biometric Security in Educational Institutions

Mission: To provide secure, efficient, and technology-enabled educational environments that prevent attendance fraud and ensure transparent communication and visibility between institutions and parents through advanced biometric security solutions.

Vision: To become the leading educational institution in Singapore recognised for implementing cutting-edge biometric security technology that ensures accurate student tracking, eliminates attendance manipulation, and provides real-time parental engagement across all our campuses.

Core Business: BioSec operates as an educational institution specialising in biometric security integration for student management. We provide comprehensive education services across multiple campuses while pioneering the use of biometric technology to ensure student safety, accurate attendance tracking, and enhanced administrative efficiency.

2. Project Description

2.1 Project Name

Biometric Student Attendance Management System

2.2 Context

BioSec's student population and administrative complexity have grown recently as a result of our recent expansion to five new campuses. The current manual attendance system has resulted in issues such as compromised integrity, lack of parental visibility, and coordination difficulty due to this expansion. Students are frequently signing attendance forms on behalf of their absent peers, also known as "buddy signing," which causes inaccuracy in calculating attendance percentages for administrative and academic reports. These problems make it difficult for parents to get accurate information about their children's actual attendance, compromise the integrity of our attendance records, and give unfair academic advantages. In order to prevent fraud and restore the integrity of our attendance management system, we require a complete biometric solution that ensures automated attendance tracking, real-time parental notification, and integrated medical certificate administration.

This project will implement a biometric fingerprint attendance management system that eliminates buddy signing through biometric identification, provides real-time attendance notifications to parents, streamlines leave management with automated medical certificate validation through our Medical Certificate System (MCS), and generates accurate attendance reports automatically. The biometric fingerprint scanner hardware has already been procured and delivered to all five BioSec campuses.

2.3 Project Goals

1. **Goal 1 (Eliminate "Buddy Signing"):** Achieve zero buddy signing incidents by implementing biometric fingerprint authentication for 100% of attendance capture across all five campuses within 6 months of project deployment, verified through audit sampling.

2. **Goal 2 (Accurate Attendance Calculation):** Reduce attendance calculation errors to less than 0.5% and eliminate manual calculation workload by 100% through automated biometric attendance tracking and reporting by month 3 of operation.

3. **Goal 3 (Real-Time Parental Notification):** Deliver automated attendance notifications to 100% of parents within 15 minutes of each lecture entry/exit via mobile application by month 2 of deployment, enabling parents with the ability to track student presence.

4. **Goal 4 (Streamlined Medical Certificate Processing):** Implement automated MCS integration for validating all medical certificates submitted with leave applications, significantly reducing fraudulent submissions and administrative processing time by at least 95% and 70% respectively, within 4 months of system deployment.

5. **Goal 5 (Multi-Campus System Integration):** Deploy unified biometric attendance system across all five campuses enabling centralised reporting and consistent student attendance, with 100% campus coverage achieved within 6 months of project start.

3. Requirements Tree

The requirements in this Project Specification are structured using a Requirements Tree approach to support clarity, completeness, and traceability through controlled granularisation of requirements. Granularisation is applied to progressively decompose high-level and abstract requirements into more specific and bounded requirements, while maintaining a clear containment relationship between levels.

In this specification, Level 0 requirements define the primary system access platforms and technical execution environments that establish the overall scope and boundaries of the system. These represent where the system is accessed or executed, such as mobile applications, web applications, attendance capture platforms, and backend hosting environments.

Level 1 requirements apply further granularisation within each Level 0 platform. For Business Operations Requirements, Level 1 expresses the major business capabilities supported by each platform. For Technical Requirements, Level 1 expresses the key technical elements, controls, and constraints contained within each platform. This approach ensures that requirements are stated at an appropriate level of abstraction and avoids premature specification of implementation details.

4. Business Operations Requirements

4.1 Level 0 – System Access Platform

BOR-0-01: Mobile Application (Essential)

Requirement: The system shall be accessible to parents and teachers through a mobile application installed on a mobile device.
Assumptions: It is assumed that parents and teachers possess a mobile device capable of running applications and have internet connectivity to access the system.
Constraints: Access to the mobile application is limited to attendance-related activities only, and users must authenticate before accessing any attendance information.
Business Rules / Regulations: Access to personal attendance data shall comply with the Personal Data Protection Act (PDPA), and users shall only be permitted to view attendance information relevant to their assigned role.

BOR-0-02: Web Application (Essential)

Requirement: The system shall be accessible to authorised administrative staff through a web-based application accessible via a designated system web address.
Assumptions: It is assumed that administrative staff have access to a computer device with network connectivity to the system.
Constraints: Access to the web application is restricted to authorised administrative roles, and the platform shall support attendance administration activities only.
Business Rules / Regulations: Administrative access to student attendance data shall comply with institutional governance policies and PDPA requirements.

BOR-0-03: Attendance Capture (Client-Server) (Essential)

Requirement: The system shall support an attendance capture platform for recording student presence during scheduled academic sessions.
Assumptions: It is assumed that biometric fingerprint scanner devices are already procured and deployed by the institution, that attendance capture occurs at designated institutional locations, and that selected scanner units may be provided to the vendor for testing purposes.
Constraints: Attendance capture is limited to officially scheduled academic sessions and is restricted to designated attendance capture points.

Business Rules / Regulations: Attendance records shall represent actual student presence, and manual attendance recording is not permitted.

4.2 Level 1 – Business Capabilities Within the System

4.2.1 Level 1 (Mobile Application – Parents & Teachers)

BOR-1-MOB-01: User Authentication Capability (Essential)

Requirement: The system shall enable users to authenticate before accessing attendance-related capabilities.
Constraints: Authentication is required before any attendance information is displayed.
Business Rules / Regulations: Authentication processes shall comply with institutional access control policies, and personal data access shall comply with PDPA requirements.

BOR-1-MOB-02: Attendance Information Access (Essential)

Requirement: The system shall provide access to attendance-related information relevant to the authenticated user.
Constraints: Parents are restricted to viewing attendance information for their child only, while teachers are restricted to viewing attendance information for their assigned classes.
Business Rules / Regulations: Attendance information presented shall be accurate and up to date.

BOR-1-MOB-03: Leave Submission Capability (Essential)

Requirement: The system shall enable parents to submit attendance-related leave information.
Constraints: Leave submissions are limited to attendance-related absence requests only.
Business Rules / Regulations: Leave requests supported by medical certificates shall be subject to validation via the Medical Certificate System (MCS), and all leave submissions shall be recorded for audit purposes.

BOR-1-MOB-04: Attendance Notifications Access (Essential)

Requirement: The system shall make attendance-related notifications available to users.
Constraints: Notifications are limited to attendance-related matters.
Business Rules / Regulations: Notifications shall comply with institutional communication policies.

4.2.2 Level 1 (Web Application – Administrators)

BOR-1-WEB-01: Attendance Oversight Capability (Essential)

Requirement: The system shall provide administrators with access to attendance status and compliance information.
Constraints: Attendance views are limited to authorised administrative roles only.
Business Rules / Regulations: Attendance compliance thresholds are defined by institutional policy, and attendance warnings shall be issued when defined thresholds are breached.

BOR-1-WEB-02: Leave and Medical Certificate Review Capability (Essential)

Requirement: The system shall enable administrators to review attendance-related leave submissions and medical certificate validation outcomes.
Constraints: Leave approval actions are restricted to authorised administrative users.
Business Rules / Regulations: Medical certificates shall be validated through the Medical Certificate System (MCS), and leave applications requiring medical evidence shall not be approved without successful validation.

4.2.3 Level 1 (Attendance Capture Platform)

BOR-1-ATT-01: Attendance Capture Capability (Essential)

Requirement: The system shall enable attendance capture during scheduled academic sessions.
Constraints: Attendance capture may only be initiated during scheduled academic sessions.
Business Rules / Regulations: Only successful biometric capture outcomes shall be recognised as valid attendance records.

BOR-1-ATT-02: Attendance Capture Outcome Indication (Essential)

Requirement: The system shall provide an indication of the outcome of attendance capture attempts.
Constraints: Outcome displays are limited to confirmation or rejection of attendance capture attempts.
Business Rules / Regulations: Attendance outcomes shall be recorded to support audit and review activities.

5. Technical Requirements (Software Configuration & IT Environment)

5.1 Level 0 – Technical Execution Platforms

TR-0-01: Mobile Application Platform

Clause: The system shall implement native mobile applications for Android and iOS platforms to support biometric attendance capture and real-time notifications for parents and teachers.
Assumptions: Parents and teachers possess compatible mobile devices and have periodic network connectivity.
Constraints: Mobile applications must support offline data caching and synchronize with cloud backend when network connectivity is restored.
Hardware:

- Android and iOS smartphones for parents and teachers
- Biometric fingerprint scanners at each campus location for student identification

Software:

- **Mobile Development:** Android Studio (Java/Kotlin), Xcode (Swift)
- **Push Notifications:** Firebase Cloud Messaging (Android), Apple Push Notification Service (iOS)
- **Local Data Storage:** SQLite for offline caching of attendance records and student information
- **Backend Integration**: RESTful API connections to Azure-hosted backend services

TR-0-02: Web-Based Administrative Platform

Clause: The system shall implement a web-based administrative interface with cloud-hosted backend for centralized management, reporting, and system configuration.

Constraints: Web portal accessible only via campus network or secure VPN connection for security purposes.

Assumptions: Schools possess compatible laptop/desktop devices and have periodic network connectivity

Hardware:

- Windows 10+ desktop or laptop computers for administrative users
- Centralized Microsoft Azure cloud infrastructure with automated scaling and load balancing across five campuses
- Network infrastructure supporting secure VPN connections for remote administrative access

Software:

- **Server Framework:** ASP.NET Core hosted on Azure App Service
- **Database:** Azure SQL Database for student records, attendance data, and system configurations
- **Web Browsers:** Chrome, Edge, Safari
- **Development Tools:** Visual Studio, Azure Data Studio
- **Server-Side Languages:** C#, T-SQL

- **Architecture:** Client-server architecture with RESTful API layer connecting web portal to cloud backend

5.2 Level 1 – Backend Infrastructure

TR-1-BE-01: Hosting & Scalability Controls

- The system shall be hosted on Microsoft Azure cloud infrastructure within the Singapore region. The backend hosting platform shall support scalable operation to support minimum 1000+ concurrent users and 200+ concurrent biometric scans.
- It should also accommodate peak usage periods (0800h-0930h). Availability expectations shall align with institutional operational requirements.

TR-1-BE-02: External System Integration

- The system shall support integration with the Medical Certificate System (MCS).
- Integration failures shall not invalidate existing attendance or leave records

TR-1-BE-03: Data Security Management

- The system shall implement security controls to protect biometric data and personal information.
- Administrative access shall enforce two-factor authentication and account lockout after five failed login attempts.
- Audit events shall be logged and retained for a minimum of twelve months.
- All handling of personal and biometric data shall comply with PDPA.

TR-1-BE-04: Performance Standards

The system shall meet defined performance benchmarks, including:

- Biometric scan processing under 2 seconds
- Web page response under 3 seconds
- API responses where 95% complete within 500ms
- Attendance reports generated within 10–60 seconds depending on size
- Biometric false reject rate shall be less than 1%, and false accept rate shall be less than 0.001%.

TR-1-BE-05: Central Database

The system shall include SQL Server database with complete schema and stored procedures. Ensure encryption for biometric data. Daily automated backups with monthly verification.

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5.3 Level 1 – Cross-Platform Requirements

TR-1-CPR-01: Data Security Management

- The system shall implement security controls to protect biometric data and personal information.
- Administrative access shall enforce two-factor authentication and account lockout after five failed login attempts.
- Audit events shall be logged and retained for a minimum of twelve months.
- All handling of personal and biometric data shall comply with PDPA.
- Mobile and web platforms shall enforce authentication and role-based access control.
- Session handling shall reduce unauthorised access risks.

TR-1-CPR-02: Performance Standards

The system shall meet defined performance benchmarks, including:

- Biometric scan processing under 2 seconds
- Web page response under 3 seconds
- API responses where 95% complete within 500ms
- Attendance reports generated within 10–60 seconds depending on size
- Biometric false reject rate (FRR) shall be less than 1%, and false accept rate (FAR) shall be less than 0.001%.

TR-1-CPR-03: Data Migration

- The system shall support migration of legacy attendance data from Excel and CSV formats.
- Current academic year data shall be migrated before system go-live.
- Duplicate Student IDs shall be detected and prevented.

TR-1-CPR-04: Maintenance & Support

- The system shall include a twelve-month maintenance warranty covering bug fixes and security patches.
- Critical system issues shall receive a response within four hours and resolution within twenty-four hours.
- Support for new mobile OS versions shall be provided within thirty days of release.

5.4 Level 1 – Mobile & Web Platforms

TR-1-CL-01: Mobile Applications

- The system shall include native Android and iOS applications registered as institutional software assets.
- Distribution of applications will be done via Google Play Store and Apple App Store.
- Offline capability with automatic sync required.
- Applications shall be code-signed with institutional certificates.
- Version updates shall be backward compatible during warranty period.
- Local data shall be encrypted.

TR-1-CL-02: Web Administration Portal

- The system shall include a web-based portal for administrators.
- Portal shall enforce role-based access control.
- Administrative access shall be restricted to campus network or secure VPN connection.

5.4 Level 1 – Documentation & Technical Artefacts

TR-1-DOC-01: Source Code Documentation

- The system shall include complete source code with inline comments and documentation headers.
- All source code shall be delivered as institutional property and documented for future maintenance

TR-1-DOC-02: Database & API Documentation

- The system shall include database schema documentation and API documentation, including request/response formats and error handling.on Items (BOR-T-08 to 11), the project budget

TR-1-DOC-03: User Manuals

- The system shall include user manuals for administrators, teachers, and parents.
- Manuals shall include screenshots and campus-specific procedures.

TR-1-DOC-04: System Architecture & Deployment Documentation

- The system shall include system architecture diagrams and deployment documentation.
- Documentation shall support future campus expansion and include disaster recovery procedures.

6. Project Financial Requirements

6.1 Cost Structure Overview

To ensure technical compliance with the Software Configuration Items (TR-1CPR-01 to 04), the project budget is distributed across eight core cost centres. This structure ensures that critical high-risk areas, such as biometric integrity and MCS integration, have sufficient funding for rigorous testing and validation. Table A.1 (Appendix A) contains a complete breakdown of these costs.

- RE & Requirement Capture (15%): Includes a 4-week intensive elicitation phase across five campuses to define the Visible Application Components (Level 1 BOR).
- Infrastructure and security (25%): Costs associated with provisioning the Microsoft Azure Singapore Region environment, such as automated scaling (TR-1-BE-01) and encryption for biometric data at rest.
- Full-Stack Development (40%): Created native codebases for Android and iOS (Java, Kotlin, and Swift) as well as the ASP.NET Core Web Portal (TR-1-BE-01).
- Integration and Migration (15%): Developed the REST API for the Medical Certificate System (MCS) and the legacy Excel data ingestion utility.
- Quality assurance and auditing (5%): Performance benchmarking (Scan < 2s) and PDPA compliance auditing.

6.2 Funding Source

BioSec's campus expansion will be funded through the capital expenditure (CAPEX) budget.

- Hardware Assumption: Biometric fingerprint scanners are provided free of charge by the institution (BOR-0-03).
- Software licensing: The budget includes Firebase Cloud Messaging (FCM) and Apple Push Notification (APNs) implementation for the 15-minute notification rule.

6.3 Payment Schedule

Payments are designed as a "Control Mechanism" to reduce the risk of Originality and Drift, one of the four horsemen of the requirements apocalypse. BioSec maintains financial leverage throughout the 13-week lifecycle by tying 70% of the total contract value to software configuration item (SCI) and external integration validation.

A 10% Retention Fund is established specifically to ensure the transition from "Project State" to "Operational State." This fund is only released after the Technical Artefact Suite is verified and delivered and the 12-month proactive maintenance period is successfully activated. See Table B.1 (Appendix B) for the full payment schedule.

6.4 Cost Optimization Considerations

The project budget is optimised by strictly adhering to the "Attendance Only" mandate, preventing "Gold Plating" (adding unnecessary features), which frequently results in software failure.

- Infrastructure Consolidation (TR-0-02 & TR-0-03): By using a single-tenant centralised Azure backend (TR-0-02), BioSec saves money on maintaining five separate server instances. Horizontal scaling (TR-1-BE-01) ensures that we only pay for high-performance computers during peak scan times (0800h–0930h).
- Hardware Lifecycle Extension (TR-1-BE-01): The system is designed to work with current COTS fingerprint scanners. This avoids a $45,000 capital expenditure on new biometric hardware.
- Strategic De-scoping (BOR-1-MOB-03): To maximise the budget for Attendance Integrity, all academic grading, appointment booking, and complaint modules were removed. This reduces the testing surface area by 30% while focussing developer time (Matching Performance).
- Automated Validation (TR-1-BE-02): By automating Medical Certificate validation through the MCS API, the administrative overhead of manual verification is reduced by approximately 15 hours per week per campus.

7. Delivery Requirements

7.1 Delivery Strategy

The project is organised into non-overlapping phases using the Waterfall Model. The strategy prioritises Attendance Integrity (BOR-1-ATT-01), which ensures that the biometric matching engine works properly before data is populated into the parental notification interfaces.

7.2 Major Delivery Milestones

Table C.1 shows the sequence of deliverables and Figure C.2 shows the Gantt Chart.

7.4 Acceptance Criteria for Delivery Completion

Delivery is considered officially complete and the final payment milestone (M5/M6) triggered only when the following criteria are met and signed off on by the BioSec Project Steering Committee:

1. **Platform and Infrastructure Operationality (TR-0-01 and TR-0-02)**
    - Multi-Campus Availability: A successful 24-hour "Soft Launch" across all five campuses (TR-1-BE-01) using a single centralised Azure SQL Database to manage multi-campus data segregation (TR-1-BE-03) .
    - Cloud Uptime: Verification of 99.9% system availability over the final 14-day evaluation period (TR-1-BE-01).

- Scalability: The system must be able to handle a simulated load of 1,000 concurrent users and 200 concurrent biometric scans without exceeding the latency thresholds (TR-1-BE-01).

2. **Biometrics and Performance Standards (TR-1-BE-05)**
    - Scan Speed: Fingerprint verification from physical contact to "Attendance Outcome Display" (BOR-1-ATT-02) must take less than two seconds.
    - Accuracy: A formal audit shows a false rejection rate (FRR) of less than 1% and a false acceptance rate (FAR) of less than 0.001%.
    - Responsiveness: Web pages for administrators must load in less than 3 seconds, and 95% of API responses must be under 500ms.
3. **Integration and Business Logic Integrity (TR-1-BE-02, TR-1-CPR-01).**
    - MCS Validation: All leave requests submitted through the mobile interface (BOR-1-MOB-03) that include medical certificates must successfully trigger a REST API call to the Medical Certificate System (TR-1-BE-01).
    - Notification Latency: Automated push notifications (Firebase/APNs) for student absence must arrive on parental mobile devices within 15 minutes of the missed session (Goal 3 / TR-0-01).
    - Data Migration: Legacy Excel ingestion validated with zero errors; all Student IDs must be unique, and duplicate records must be successfully flagged/blocked (TR-1-CPR-01).
4. **Security, Privacy, and Audit (BOR-1-WEB-01 and TR-1-BE-04)**
    - Compliance: A formal data privacy audit confirms that all biometric data is encrypted at rest using AES-256 and TLS 1.2+ in transit.
    - Access Control: Ensure that Role-Based Access Control (RBAC) prevents parents from viewing data other than their own child's (BOR-1-MOB-02).
    - Audit Trail: Successfully generated a system log with a 100% trace of all attendance changes and administrative logins (TR-1-BE-04).
5. **Documentation and Technical Artefacts (BOR-T-11 and BOR-T-12)**
    - Source Code Handover: Delivery of the entire Git repository, which includes native Java/Kotlin (Android), Swift (iOS), and C# (Backend) code with English-language inline documentation (TR-1-DOC-01).
    - Executable Artefacts: Provide production-ready SQL Schema scripts and API documentation (in Swagger/OpenAPI format) (TR-1-DOC-02).
    - User Manual Suite: Created specialised manuals for three distinct roles (Administrators, Teachers, and Parents) in PDF and editable DOCX formats (TR-1-DOC-03).

8. Sign Off

This section certifies that the Android-based Biometric Student Attendance System Project Specification has been examined and approved for publication. By signing below, the Project Lead confirms that:

- The project specification has been finished and verified internally.
- Vendor response requirements and acceptance criteria are well-defined.
- The document is authorised for distribution as a component of the bidding or tender procedure.
- This approval does not signify project closure or system acceptance.

Loh Wen Xuan		30/01/2026
Name (Project Lead)	Signature	Date

9. Addendum

9.1 Appendix A: Project Cost Breakdown Table

Cost Category	**Description**	**Technical Reference**	**Cost Allocation (%)**
Requirements & System Design	Elicitation across multiple campuses (5 sites), stakeholder interviews, and definition of the Level 0/1 Access Platform.	BOR-0-01 to 03 / BOR-1	15%
Native Mobile Development	Native programming for Android (Java/Kotlin) and iOS (Swift). Implementing a biometric matching engine.	TR-T-01 / TR-T-10	25%
Web Portal & Backend Dev	ASP.NET Core Admin Portal and Azure SQL Database schema creation with multi-campus segregation.	BOR-T-02	20%
External Systems Integration	Create secure REST API links for the Medical Certificate System (MCS) and SMTP/Push notification servers.	TR-1-BE-02	15%
Data Migration & Ingestion	Creating an automated Excel/CSV ingestion tool with pre-import validation for Student IDs.	TR-1-CPR-03	5%

Quality Assurance (QA)	Biometric FRR/FAR stress testing, security vulnerability assessment (SQLi/XSS), and PDPA compliance checks.	TR-1-BE-03	5%
Deployment & Training	Production rollout across five campuses, user onboarding, and delivery of technical artefacts (manual/source code).	TR-1-DOC-01 to TR-1-DOC04	5%
Maintenance Warranty	12-month post-deployment support, monthly system health reports, and critical patch management.	TR-1-CPR-04	5%
Total			100%

Table A.1

9.2 Appendix B: Payment Schedule Table

Milestone	**Description**	**Due**	**Payment (%)**
Project Kick-off & PS Approval	Requirement Finalization: Completion of multi-campus elicitation and formal sign-off on Level 0 and Level 1 Business Operations Requirements (BOR-0/BOR-1).	1 February 2026	15%
System Architecture & UI Baseline	Technical Blueprint: Approved Azure Infrastructure Design (TR-0-02), SQL Schema (BOR-T-05), VPN/Security Configuration (TR-1-BE-03).	15 February 2026	15%
Core System Implementation	Functional Platform Delivery includes compiled Native Apps for Android and iOS (TR-1-CL-01), as well as the Web Administration Portal (TR-1-CL-02) with Role-Based Access Control.	22 March 2026	25%
Integration & System Testing	Systems connectivity includes a successful REST API handshake with the Medical	29 March 2026	20%

	Certificate System (TR-1-BE-02) and validation of performance benchmarks/scan speeds (TR-1-BE-04).		
Handover	Institutional asset delivery includes the final transfer of documented source code (TR-1-DOC-01), API/database documentation (BOR-T-13), and the User Manual Suite (TR-1-DOC-03).	5 April 2026	10%
Warranty Commencement	Retention Release: Successful activation of the 12-month Maintenance & Support period (TR-1-CPR-04) and verification of system stability during the first operational phase.	5 July 2027	10%
Total			100%

Table B.1

9.3 Appendix C: Delivery Milestones Table

Phase	Detailed Deliverable (Technical Artefacts)	Verification Method	Associated Gantt ID
Project Initiation	Stakeholder Elicitation Report: A compilation of interview data from five campuses, as well as localised site network assessments (BOR-0-01 to 03)	Signed Interview Transcripts & Site Reports	1.1
Requirements Finalisation	Baseline PS & SRS (BOR-T-06): PDPA data mapping and signed-off document locking on Level 0/1 platforms (BOR-1).	Physical/Digital Signature Page	1.2
System Design	The Architecture Blueprint (TR-1-DOC-02) includes the SQL Entity Relationship Diagram (ERD), API Documentation	Technical Peer Review Sign-off	2.1

	(TR-1-DOC-02), and an Azure Cloud configuration map (TR-1-DOC-04).		
Development	Software configuration items (TR-1-CL-01, TR-1-CL-02) include functional Android/iOS applications as well as an administrative web portal (Alpha build).	Sandbox Environment Demo	3.1
Integration & Testing	Integration Handshake: Verified REST API connection to MCS (BOR-1-WEB-02) and successful import of legacy Excel student records.	System Integration Test (SIT) Report	4.1
User Acceptance Testing	Validated UAT scripts include biometric performance benchmarking, leave submission (BOR-1-ATT-02), and notification delivery tests.	Stakeholder UAT Sign-off Matrix	5.2
Deployment	Production Environment: 100% go-live on BioSec Azure Tenant (BOR-1-ATT-01), including live biometric terminal connectivity.	Post-Implementation Audit	6.1
Warranty Support	Governance Suite: delivery of source code (TR-1-DOC-01), user manuals (TR-1-DOC-03), and the start of maintenance and support (TR-1-CPR-04).	Receipt of Technical Assets	-

Table C.1

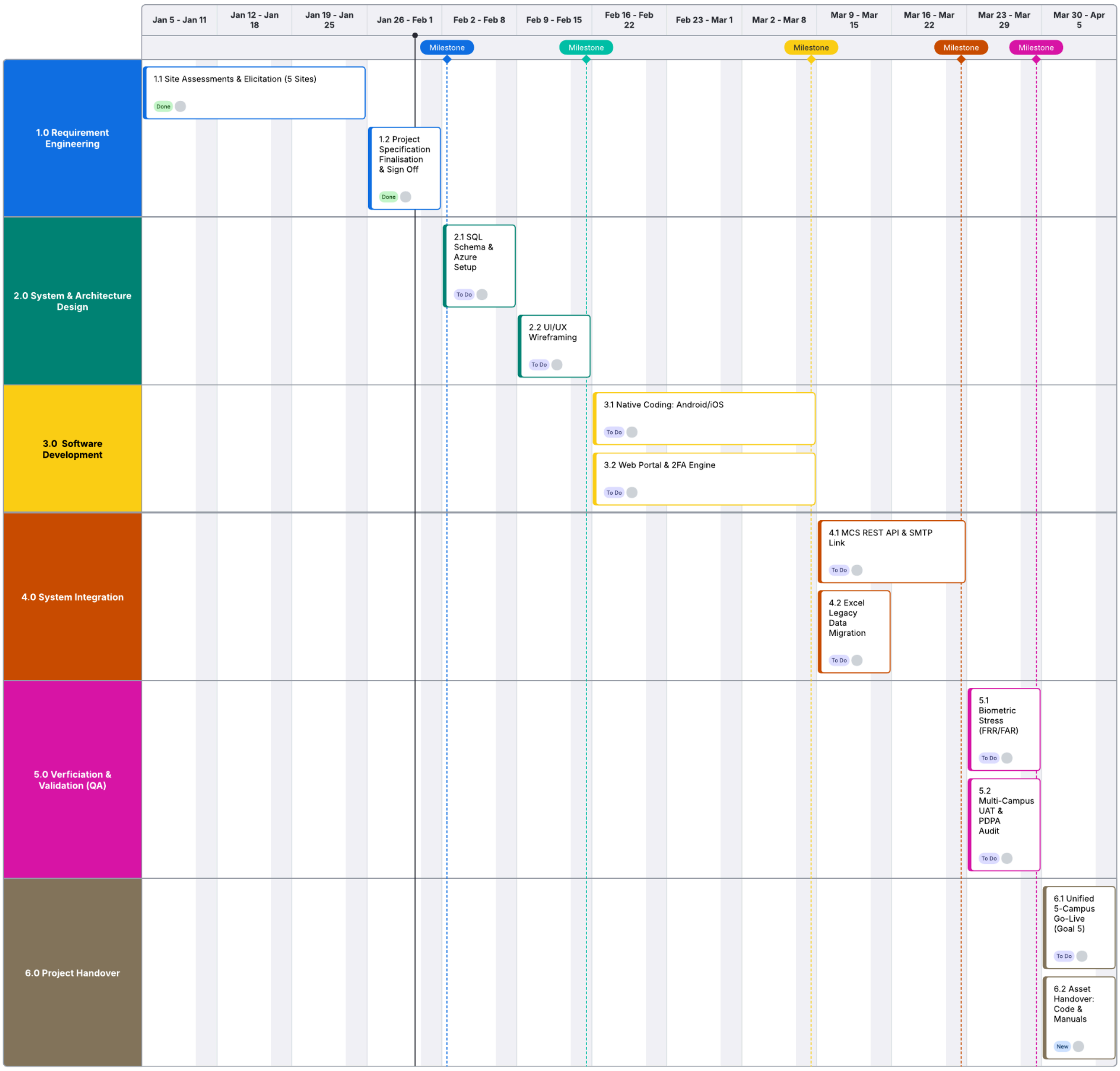
Jan 5 - Jan 11
Jan 12 - Jan 18
Jan 19 - Jan 25
Jan 26 - Feb 1
Feb 2 - Feb 8
Feb 9 - Feb 15
Feb 16 - Feb 22
Feb 23 - Mar 1
Mar 2 - Mar 8
Mar 9 - Mar 15
Mar 16 - Mar 22
Mar 23 - Mar 29
Mar 30 - Apr 5
Milestone
Milestone
Milestone
Milestone
Milestone
1.0 Requirement Engineering
1.1 Site Assessments & Elicitation (5 Sites)
Done
1.2 Project Specification Finalisation & Sign Off
Done
2.0 System & Architecture Design
2.1 SQL Schema & Azure Setup
To Do
2.2 UI/UX Wireframing
To Do
3.0 Software Development
3.1 Native Coding: Android/iOS
To Do
3.2 Web Portal & 2FA Engine
To Do
4.0 System Integration
4.1 MCS REST API & SMTP Link
To Do
4.2 Excel Legacy Data Migration
To Do
5.0 Verficiation & Validation (QA)
5.1 Biometric Stress (FRR/FAR)
To Do
5.2 Multi-Campus UAT & PDPA Audit
To Do
6.0 Project Handover
6.1 Unified 5-Campus Go-Live (Goal 5)
To Do
6.2 Asset Handover: Code & Manuals
New

Figure C.2

9.4 Appendix D: Requirements Modelling

9.4.1 Data Dictionary

UserAccount

Field name	Data type	Size	Allow null?	Description	Example
user_id	INT	11	No	PK: Unique account ID	1001
username	VARCHAR	50	No	Login username	admin1
password_hash	VARCHAR	256	No	Encrypted password	(SHA256 hash)
role	ENUM	–	No	'Admin', 'Parent', 'Teacher'	'Parent'
email	VARCHAR	100	No	Registered email for notifications	parent@school.sg
created_date	DATETIME	–	No	Account creation timestamp	2026-01-01 10:00:00

Table D.1

Student

Field name	Data type	Size	Allow null?	Description	Example
student_id	INT	11	No	PK: Unique student ID	2026001
first_name	VARCHAR	50	No	Given name	Ah
last_name	VARCHAR	50	No	Family name	Mei
date_of_birth	DATE	–	No	Birth date	2010-05-15
fingerprint_hash	VARCHAR	256	No	Biometric hash for attendance matching	(fingerprint hash)
parent_id	INT	11	No	FK to UserAccount (parent account)	2001
status	VARCHAR	20	Yes	Active, Inactive, Suspended	Active

Table D.2

Teacher

Field name	Data type	Size	Allow null?	Description	Example
teacher_id	INT	11	No	PK: Unique teacher ID	T001
first_name	VARCHAR	50	No	Given name	Mr. Tan
last_name	VARCHAR	50	No	Family name	Wei
user_id	INT	11	No	FK to UserAccount	3001
contact_phone	VARCHAR	15	Yes	Phone for notifications	+6581234567

Table D.3

AttendanceRecord

Field name	Data type	Size	Allow null?	Description	Example
attendance_id	INT	11	No	PK: Unique record ID	500001
student_id	INT	11	No	FK to Student	2026001
teacher_id	INT	11	No	FK to Teacher (who took attendance)	T001
lecture_datetime	DATETIME	–	No	Scheduled lecture start	2026-02-03 09:00:00
entry_time	DATETIME	–	Yes	Biometric entry scan time	2026-02-03 09:01:30
exit_time	DATETIME	–	Yes	Biometric exit scan time	2026-02-03 10:25:45
status	ENUM	–	No	Present, Late, Absent, Excused	Present
leave_id	INT	11	Yes	FK to LeaveRequest (if Excused)	7001

Table D.4

LeaveRequest

Field name	Data type	Size	Allow null?	Description	Example
leave_id	INT	11	No	PK: Unique leave request ID	7001
student_id	INT	11	No	FK to Student	2026001
parent_id	INT	11	No	FK to UserAccount (requester)	2001
start_date	DATE	–	No	Leave start date	2026-02-05
end_date	DATE	–	No	Leave end date	2026-02-07
reason	TEXT	–	No	Brief reason (e.g., illness)	Sick leave
mc_reference	VARCHAR	100	Yes	MCS system reference for medical cert	MCS-2026-001
status	ENUM	–	No	Pending, Approved, Rejected, Cancelled	Approved
admin_note	TEXT	–	Yes	Admin response/comments	Approved with MC
request_date	DATETIME	–	No	Timestamp of request	2026-02-04 14:30:00

Table D.5

WarningLetter

Field name	Data type	Size	Allow null?	Description	Example
warning_id	INT	11	No	PK: Unique warning ID	8001
student_id	INT	11	No	FK to Student	2026001
parent_id	INT	11	No	FK to UserAccount (recipient)	2001
issue_date	DATE	–	No	Date warning issued	2026-02-10
attendance_pct	DECIMAL	5,2	No	Attendance percentage triggering warning	75.50
message	TEXT	–	No	Warning content (attendance <80%)	Improve attendance
parent_reply	TEXT	–	Yes	Parent response	Will monitor

Table D.6

Notification

Field name	Data type	Size	Allow null?	Description	Example
notif_id	INT	11	No	PK: Unique notification ID	9001
user_id	INT	11	No	FK to UserAccount (recipient)	2001
type	VARCHAR	30	No	'LeaveApproved', 'LeaveRejected', 'Warning'	LeaveApproved
title	VARCHAR	100	No	Short title	Leave Approved
message	TEXT	–	No	Full message body	Your child's leave...
related_id	INT	11	Yes	FK to related entity (e.g., leave_id)	7001
sent_datetime	DATETIME	–	No	Sent timestamp	2026-02-04 15:00:00
read_status	BOOLEAN	–	No	Read or unread	0 (unread)

Table D.7

AdminLog

Field name	Data type	Size	Allow null?	Description	Example
log_id	INT	11	No	PK: Unique log entry ID	10001
admin_user_id	INT	11	No	FK to UserAccount (admin who logged)	1001
student_id	INT	11	No	FK to Student	2026001
action_type	VARCHAR	20	No	'ManualEntry', 'ManualExit', 'Override'	ManualEntry
timestamp	DATETIME	–	No	Action time	2026-02-03 09:05:00
notes	TEXT	–	Yes	Reason for manual action	Scanner fault

Table D.8

ReportView

Field name	Data type	Size	Allow null?	Description	Example
report_id	INT	11	No	Generated report ID	R001
student_id	INT	11	No	FK to Student	2026001
period_start	DATE	–	No	Report period start	2025-12-01
period_end	DATE	–	No	Report period end	2026-01-31
total_lectures	INT	6	No	Total lectures in period	45
present_count	INT	6	No	Days present	38
attendance_pct	DECIMAL	5,2	No	Calculated percentage	84.44

Table D.9

9.4.2 DFD Context Diagram (Level 0)

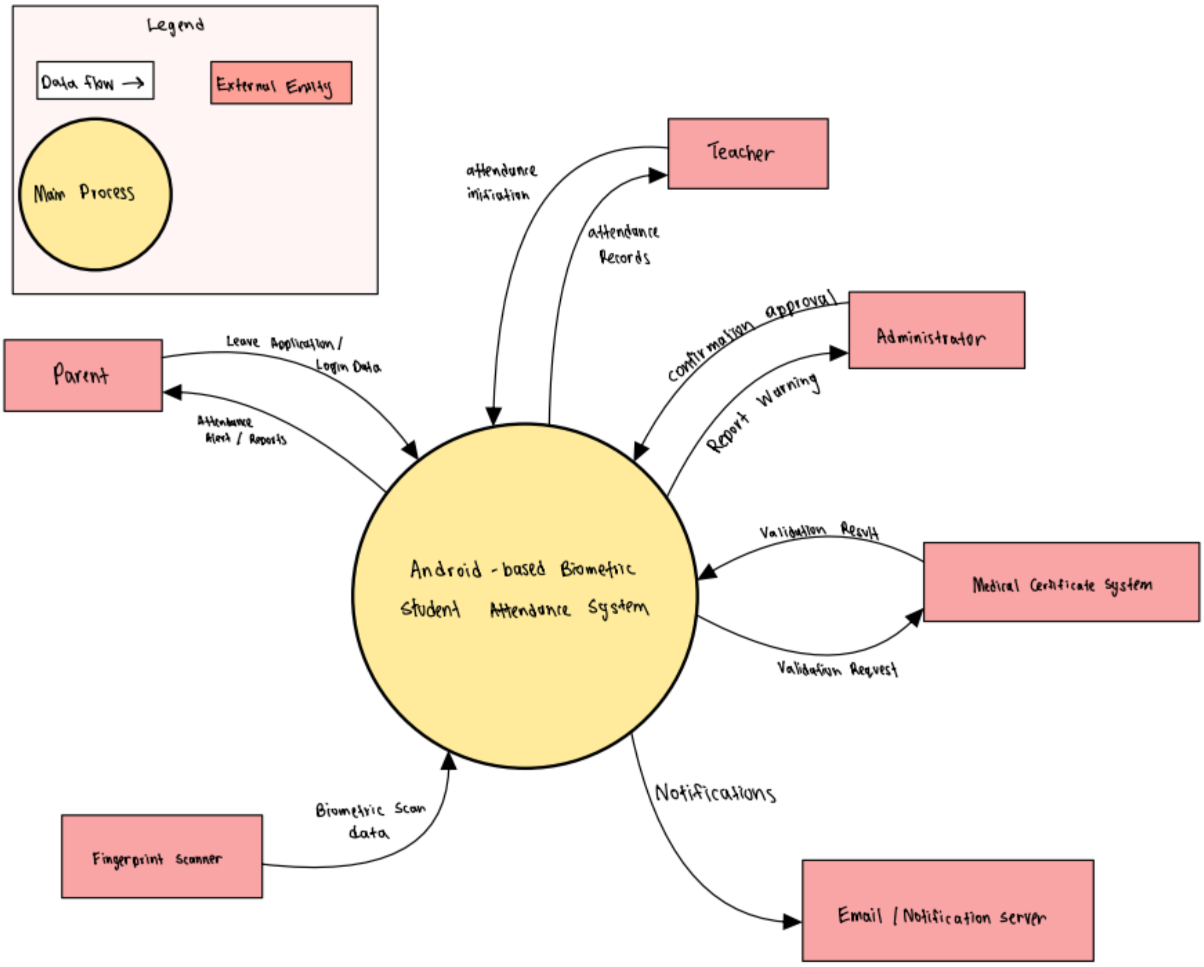

Diagram D.10